



STATS/CSE 780

Assignment 3

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Introduction

The following dataset[1] aims to develop a customer segmentation analysis to define an effective marketing strategy for a credit card company. The data set used in this analysis summarizes the usage behavior of approximately 9000 active credit card holders during the past six months. The dataset is available at the customer level and comprises 18 behavioral variables that provide valuable insights into customer behavior. The behavioral variables in the dataset include factors such as transaction frequency, transaction amount, balance, credit limit, and payment amount. Other variables include demographic information such as age, gender, and income, as well as variables that describe customer behavior, such as tenure, loyalty, and credit card type. The customer segmentation analysis will help the credit card company divide its customer base into distinct groups with similar characteristics, needs, and behaviors. This analysis will provide insights into the different groups' preferences, enabling the company to develop a targeted marketing strategy that is tailored to each group's specific needs.

Data Transformation and Preprocessing

Handling missing values

There are 1690 missing values(NA) which will be removed from the data set:

Outliers analysis

According to the box plot in the supplementary document, there are some outliers which can be handled by IQR(Inter Quartaile Range) technique:

In order to select important variables,pricipal components analysis is used as follows:

Cluster Analysis

K-means Clustering

Determine the number of clusters

In order to use effective clustering, firstly the optimal number of clusters has been extracted by using Elbow method width method as follows:

Elbow method

According to the Elbow and Silhouette the optimal number of clusters is $K=3$. In the following section, $k=3$ considered as a cluster numbers.

Get Average silhouette width

Seems relative good clustering with kmeans.

Hierarchical Clustering

In order to use effective Hierarchical Clustering clustering, firstly the optimal number of clusters has been extracted by using Elbow method width method as follows

Elbow method for hiercahical clustering

the best cutoff in this clustering is 4.

Again the results of hierarchical is fairly good. clustering with

Pricipal Components Analysis

Firstly the number of principle components are extracted by Elbow Methods: As It can be seen two principle components is acceptableness for the next combination by other techniques:

Principal Components analysis and Hierarchical

By Combining Principal Components analysis and Hierarchical, silhouette score is score remain stable.

Visualize Clustering Results of using PCA and KMeans

After applying PCA, the quality of K-Means has been dramatically increased to 0.53 Silhouette average score

Comparison with different clustering algorithms

Comparison 1 : According to the obtained silhouette average between K-means and hierarchical, hierarchical clustering has better results (Silhouette average:0.56) than K-Means with (Silhouette average:0.56).

Comparison 2: After applying PCA, the quality of K-Means has been dramatically increased to 0.53 Silhouette average score. However,the mixture of Hierarchical and PCA performance remains stable.

References

1. <https://archive.ics.uci.edu/ml/datasets/bank+marketing>



STATS/CSE 780
Technical Supplemental

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```
# Load Bank Marketing Archive dataset
```

```
original_data<- read_csv("CC_GENERAL.csv",show_col_types = FALSE)
```

```
# Summary of Data
```

```
dim(original_data)
```

```
## [1] 8950 18
```

```
#summary(original_data)
```

```
cat("Number of missing values is",sum(is.na(original_data)),  
    "\n")
```

```
## Number of missing values is 314
```

```
clean_data <- na.omit(original_data)
```

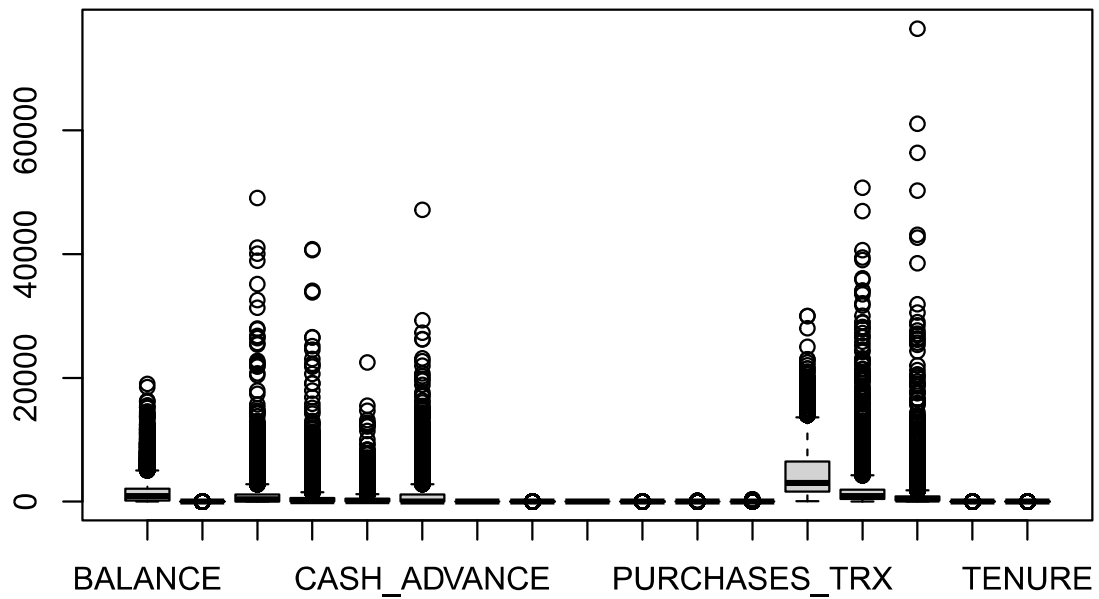
```
# check the missing values
```

```
sum(is.na(clean_data))
```

```
## [1] 0
```

```
clean_data <-clean_data[,-1]
```

```
boxplot(clean_data)
```



```
# Extract column names
c <- colnames(original_data[,-1])

# Define function for calculating IQR and removing outliers
IQRr <- function(x){
  Q1 <- quantile(x, 0.25)
  Q3 <- quantile(x, 0.75)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
  upper_bound <- Q3 + 1.5 * IQR
  x[x < lower_bound] <- NA
  x[x > upper_bound] <- NA
  return(x)
}
```

```

}

# Apply function to all columns
iqr_list <- lapply(clean_data[, c], IQR)
iqr_df <- as.data.frame(iqr_list)
clean_data <- na.omit(iqr_df)

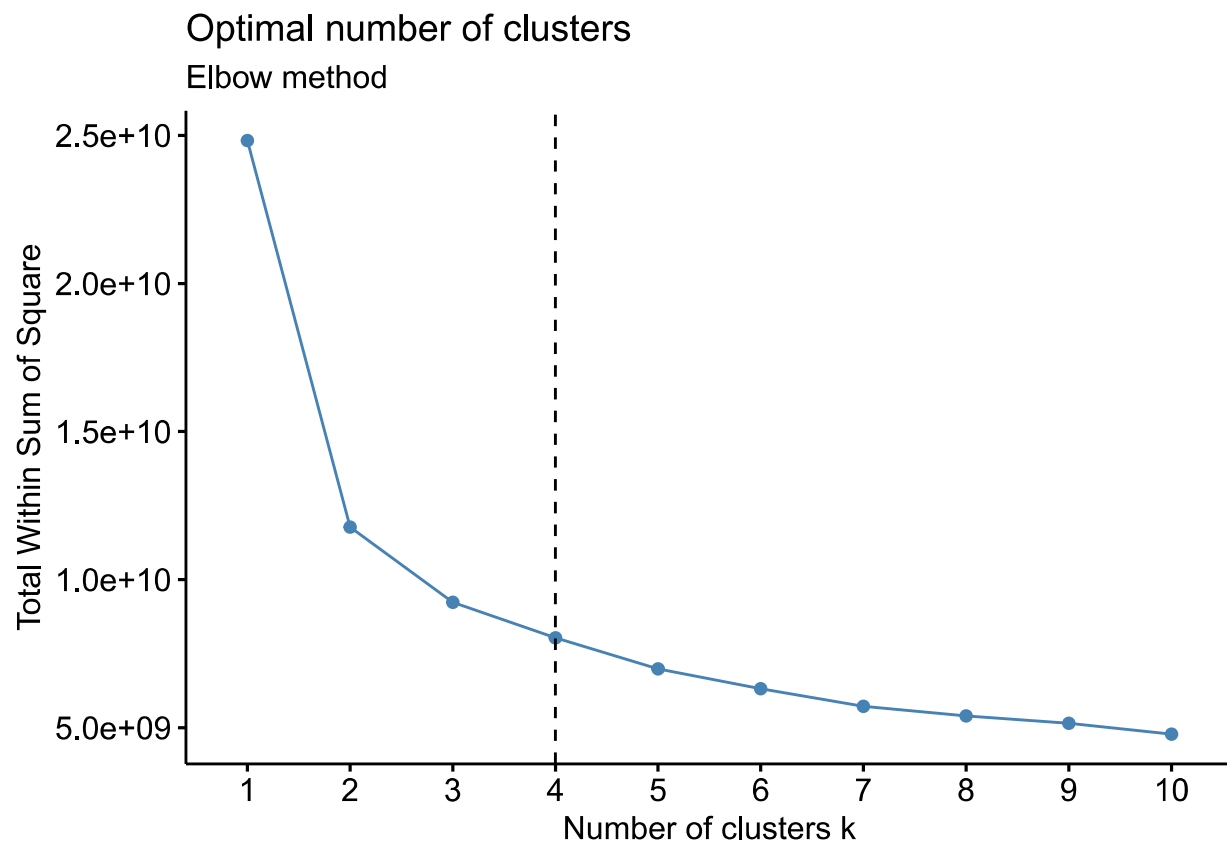
```

Elbow method

```

fviz_nbclust(clean_data, FUNcluster = kmeans, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2) +
  labs(subtitle = "Elbow method")

```

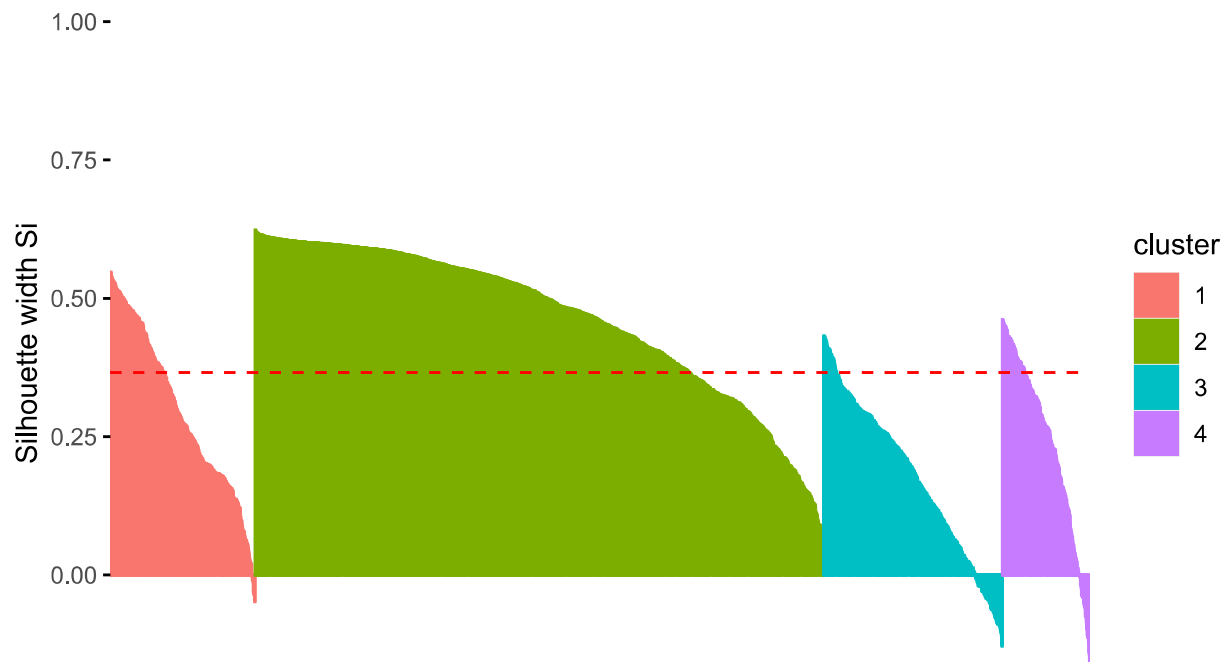


```
set.seed(2)
clustering <- kmeans(clean_data, centers = 4, nstart = 20)
```

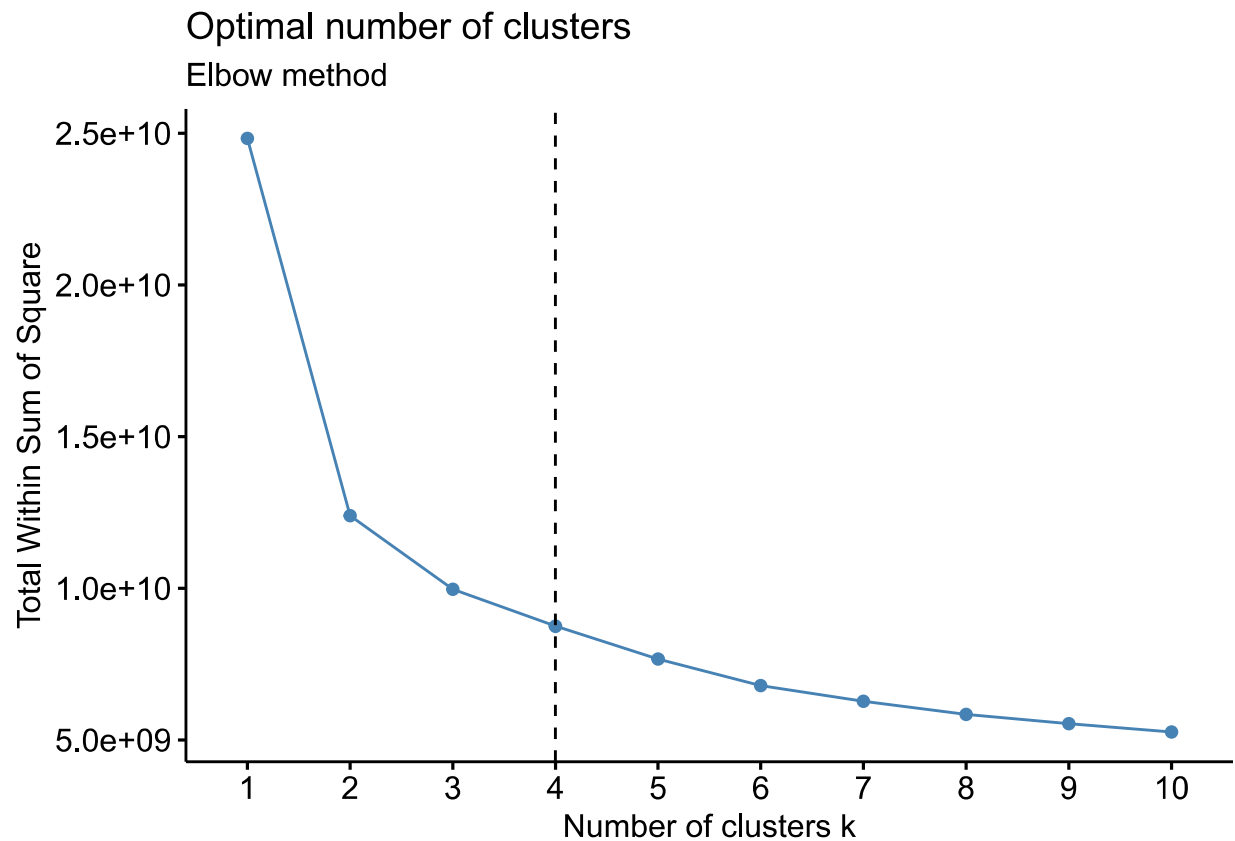
```
sil <- silhouette(clustering$cluster, dist(clean_data))
fviz_silhouette(sil)
```

```
##   cluster size ave.sil.width
## 1         1  436         0.29
## 2         2 1714         0.46
## 3         3  538         0.17
## 4         4  263         0.24
```

Clusters silhouette plot
Average silhouette width: 0.37



```
fviz_nbclust(clean_data, FUNcluster = hcut, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2) +
  labs(subtitle = "Elbow method")
```



the best cutoff in this clustering is 4.

```
# Set cluster number
m <- 4

# Set cluster name
c1 <- "cluster1"
c2 <- "cluster2"
c3 <- "cluster3"
c4 <- "cluster4"

# Perform hierarchical clustering
cluster <- clean_data %>%
  dist(method = "euclidean") %>%
  hclust(method = "ward.D")
```

```

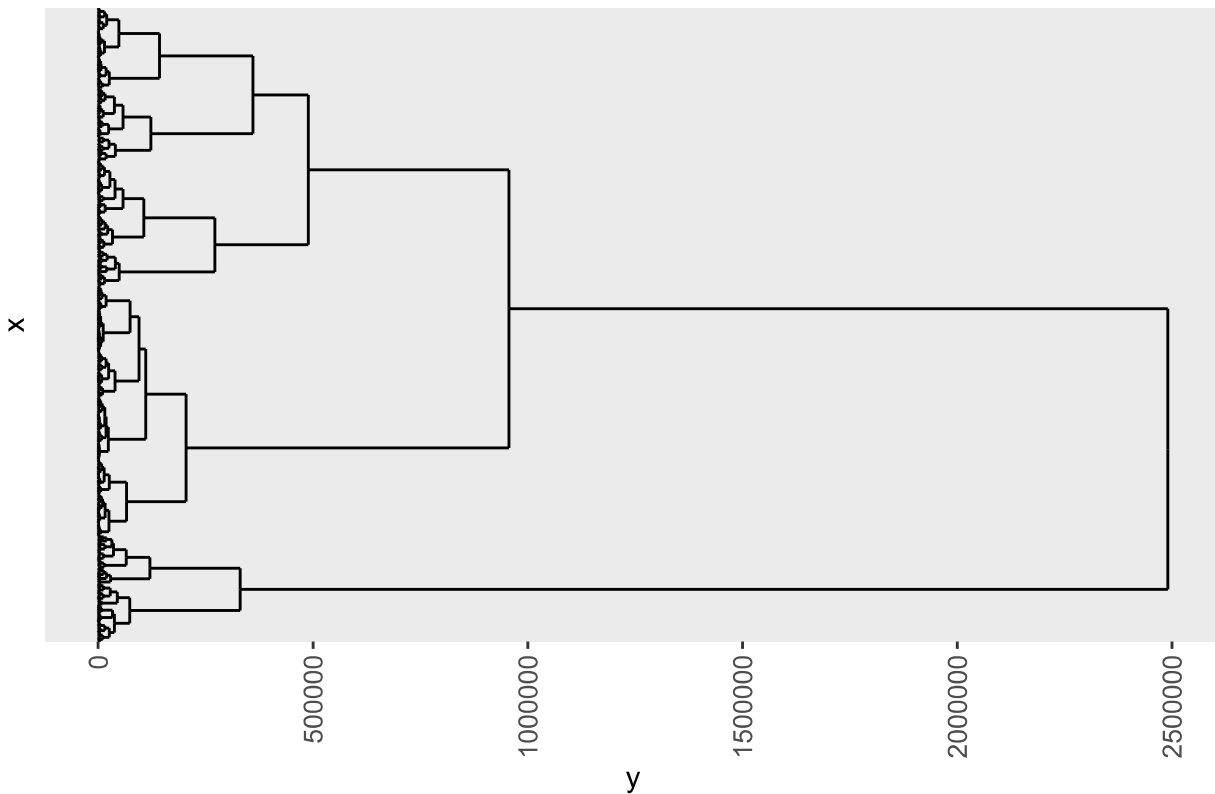
# Plot dendrogram
ggdendrogram(cluster, rotate = TRUE, theme_dendro = FALSE) +
  labs(title = paste0("Hierarchical clustering with ", m, " clusters")) +
  scale_x_continuous(expand = c(0, 0)) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank(),
        plot.title = element_text(size = 14, face = "bold"),
        axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 10))

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Hierarchical clustering with 4 clusters



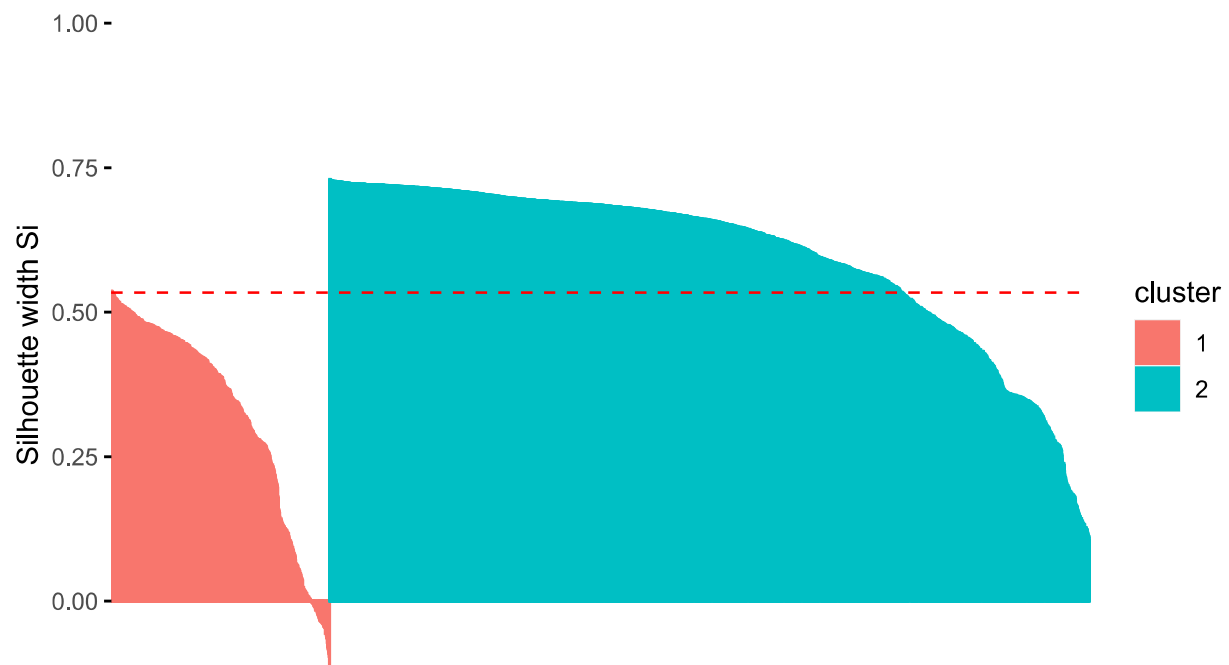
```
pca.out <- prcomp(clean_data, scale. = FALSE)

set.seed(123)

km.out <- kmeans(pca.out$x[,1:2], 2, nstart = 20)
km_pca <- km.out
sil <- silhouette(km_pca$cluster, dist(clean_data))
fviz_silhouette(sil)
```

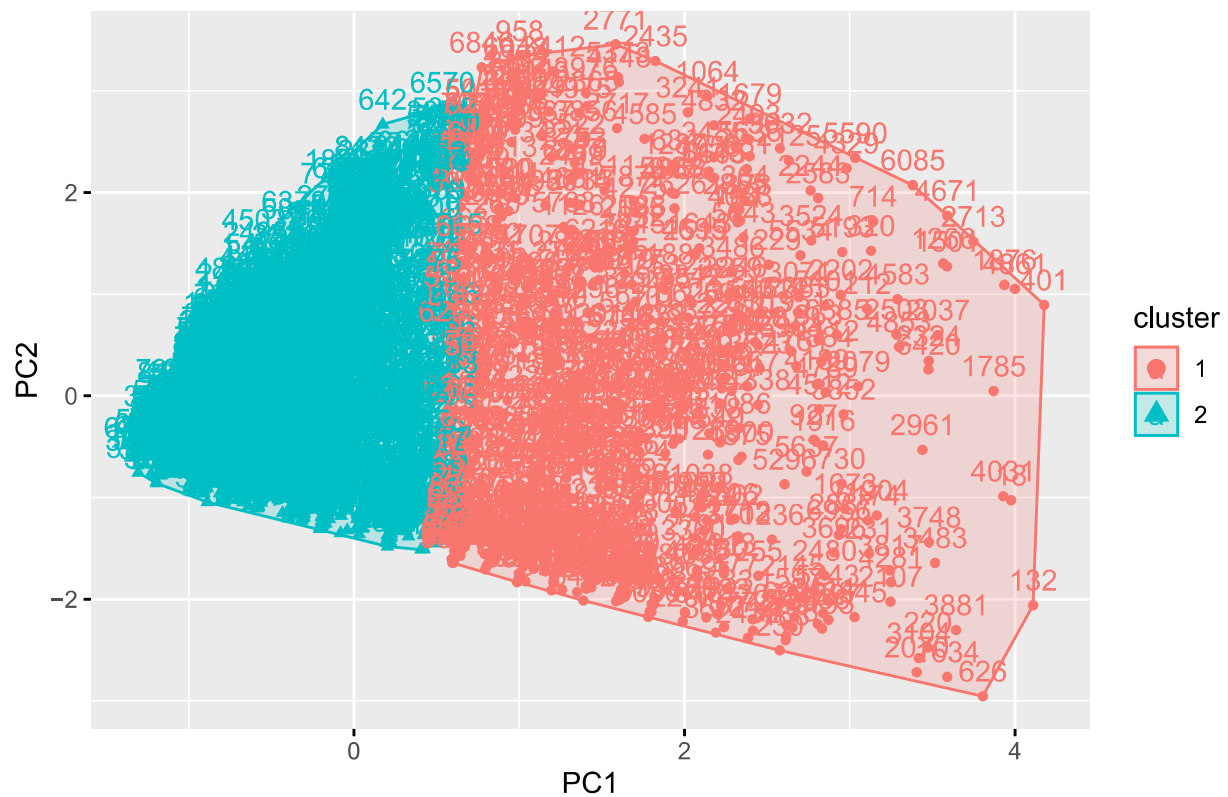
```
##   cluster size ave.sil.width
## 1         1  658          0.32
## 2         2 2293          0.60
```

Clusters silhouette plot
Average silhouette width: 0.53



```
fviz_cluster(km.out, pca.out$x[,1:2])
```

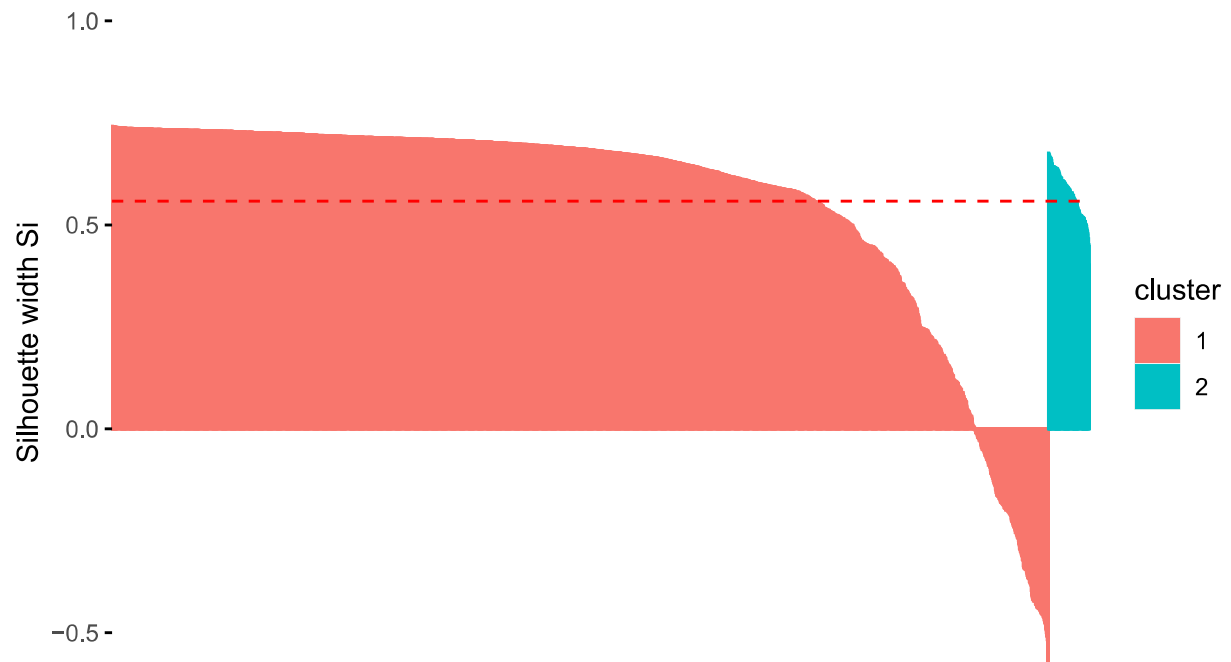
Cluster plot



```
hc.out <- prcomp(clean_data, scale. = FALSE)
hc_pca <- hclust(dist(pca.out$x[,1:2]))
hc_clusters <- cutree(hc_pca, k=2)
sil <- silhouette(hc_clusters, dist(clean_data))
fviz_silhouette(sil)
```

```
##   cluster size ave.sil.width
## 1      1 2824          0.56
## 2      2  127          0.58
```

Clusters silhouette plot
Average silhouette width: 0.56



```
fviz_cluster(list(data = pca.out$x[,1:2], cluster = hc_clusters))
```

